

Algorithm Name

Clipped Adam with Momentum (gradient-norm clipping with first and second moments). For theory, we also consider the special case with no momentum and no adaptivity (pure clipped gradient descent), which is a special case obtained by setting $\beta_1 = \beta_2 = 0$ and omitting bias-corrections.

Mathematical Formulation

We consider minimizing a differentiable (possibly non-convex) function $f : \mathbb{R}^d \rightarrow \mathbb{R}$. Let $\nabla f(x)$ denote its gradient at x .

Clipping operator (global ℓ_2 -norm clipping): for a clipping threshold $C > 0$,

$$\text{clip}_C(g) \triangleq \frac{g}{\max\{1, \|g\|_2/C\}}.$$

Given hyperparameters $\eta > 0$ (base stepsize), $\beta_1 \in [0, 1]$ (first-moment momentum), $\beta_2 \in [0, 1]$ (second-moment momentum), $\epsilon > 0$ (stabilizer), the Clipped Adam with Momentum updates are:

$$\begin{aligned} g_t &\leftarrow \nabla f(x_t), \\ u_t &\leftarrow \text{clip}_C(g_t) = \frac{g_t}{\max\{1, \|g_t\|_2/C\}}, \\ m_t &\leftarrow \beta_1 m_{t-1} + (1 - \beta_1) u_t, \\ v_t &\leftarrow \beta_2 v_{t-1} + (1 - \beta_2)(u_t \odot u_t) \quad (\text{elementwise square}), \\ \hat{m}_t &\leftarrow \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t \leftarrow \frac{v_t}{1 - \beta_2^t}, \\ x_{t+1} &\leftarrow x_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \quad (\text{division is elementwise}). \end{aligned}$$

Initialization: $x_1 \in \mathbb{R}^d$ given, $m_0 = 0 \in \mathbb{R}^d$, $v_0 = 0 \in \mathbb{R}^d$.

For analysis, we will also study the special case (Clipped Gradient Descent, CGD):

$$x_{t+1} \leftarrow x_t - \eta u_t \quad \text{with} \quad u_t = \text{clip}_C(\nabla f(x_t)),$$

which is obtained by setting $\beta_1 = \beta_2 = 0$ and dropping bias corrections.

Pseudocode

Algorithm 1 Clipped Adam with Momentum

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1: Input: initial point  $x_1 \in \mathbb{R}^d$ , stepsize  $\eta > 0$ , clipping threshold  $C > 0$ , momentum  $\beta_1 \in [0, 1)$ ,
   second-moment  $\beta_2 \in [0, 1)$ , stabilizer  $\epsilon > 0$ , max iterations  $T$ 
2: Initialize  $m_0 \leftarrow 0$ ,  $v_0 \leftarrow 0$ 
3: for  $t = 1$  to  $T$  do
4:    $g_t \leftarrow \nabla f(x_t)$ 
5:    $u_t \leftarrow g_t / \max\{1, \|g_t\|_2 / C\}$  (global  $\ell_2$ -norm clipping)
6:    $m_t \leftarrow \beta_1 m_{t-1} + (1 - \beta_1) u_t$ 
7:    $v_t \leftarrow \beta_2 v_{t-1} + (1 - \beta_2)(u_t \odot u_t)$ 
8:    $\hat{m}_t \leftarrow m_t / (1 - \beta_1^t)$ ,  $\hat{v}_t \leftarrow v_t / (1 - \beta_2^t)$ 
9:    $x_{t+1} \leftarrow x_t - \eta \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon)$  (elementwise division)
10: end for
11: Output:  $x_{T+1}$ 

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Dry Run Example

We apply the algorithm to $f(w) = (w - 3)^2$ (one dimension, $d = 1$), with $w_1 = 0$, $\eta = 0.1$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, $C = 2$, for 3 iterations. In $d = 1$, vectors are scalars and elementwise operations reduce to scalar ones.

At step t : - Gradient: $g_t = f'(w_t) = 2(w_t - 3)$ - Clipping: $u_t = g_t / \max\{1, |g_t|/2\}$

Initialization: $m_0 = 0$, $v_0 = 0$.

Iteration $t = 1$: - $w_1 = 0$, $f(w_1) = 9$, $g_1 = 2(0 - 3) = -6$, $|g_1| = 6 > C$ - $u_1 = -6/(6/2) = -2$
 - $m_1 = 0.9 \cdot 0 + 0.1 \cdot (-2) = -0.2$ - $v_1 = 0.999 \cdot 0 + 0.001 \cdot 4 = 0.004$ - Bias-corrected: $\hat{m}_1 = -0.2/(1 - 0.9) = -2$, $\hat{v}_1 = 0.004/(1 - 0.999) = 4$ - Update: $w_2 = w_1 - 0.1 \cdot (-2)/(\sqrt{4} + 10^{-8}) = 0 - 0.1 \cdot (-2)/2 = 0.1$ - $f(w_2) = (0.1 - 3)^2 = 8.41$

Iteration $t = 2$: - $w_2 = 0.1$, $g_2 = 2(0.1 - 3) = -5.8$, $|g_2| = 5.8 > C$ - $u_2 = -5.8/(5.8/2) = -2$
 - $m_2 = 0.9(-0.2) + 0.1(-2) = -0.18 - 0.2 = -0.38$ - $v_2 = 0.999 \cdot 0.004 + 0.001 \cdot 4 = 0.003996 + 0.004 = 0.007996$ - $\hat{m}_2 = -0.38/(1 - 0.9^2) = -0.38/0.19 = -2$, $\hat{v}_2 = 0.007996/(1 - 0.999^2) = 0.007996/0.001999 = 4$ - $w_3 = 0.1 - 0.1 \cdot (-2)/2 = 0.2$ - $f(w_3) = (0.2 - 3)^2 = 7.84$

Iteration $t = 3$: - $w_3 = 0.2$, $g_3 = 2(0.2 - 3) = -5.6$, $|g_3| > C$ - $u_3 = -2$ - $m_3 = 0.9(-0.38) + 0.1(-2) = -0.342 - 0.2 = -0.542$ - $v_3 = 0.999 \cdot 0.007996 + 0.001 \cdot 4 = 0.007988004 + 0.004 = 0.011988004$ - $\hat{m}_3 = -0.542/(1 - 0.9^3) = -0.542/0.271 = -2$, $\hat{v}_3 = 0.011988004/(1 - 0.999^3) = 0.011988004/0.002997001 = 4$ - $w_4 = 0.2 - 0.1 \cdot (-2)/2 = 0.3$ - $f(w_4) = (0.3 - 3)^2 = 7.29$

Assumptions

We analyze the clipped method under standard smooth non-convex assumptions.

Assumption 1 (Smoothness and boundedness). - $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is differentiable and L -smooth: for all $x, y \in \mathbb{R}^d$, $\|\nabla f(x) - \nabla f(y)\|_2 \leq L\|x - y\|_2$. - f is bounded below: $\inf_{x \in \mathbb{R}^d} f(x) \geq f_{\inf} > -\infty$.

Assumption 2 (Algorithmic parameters). - Clipping threshold $C > 0$, stepsize $\eta > 0$. - For the main theorem, we analyze the special case $\beta_1 = \beta_2 = 0$ (Clipped Gradient Descent, CGD). This is a special case of the proposed algorithm (set $\beta_1 = \beta_2 = 0$ and remove bias-corrections).

Main Convergence Theorem

Theorem 1 (Global convergence of Clipped Gradient Descent). *Let Assumptions 1–2 hold and consider CGD $x_{t+1} = x_t - \eta u_t$ with $u_t = \text{clip}_C(\nabla f(x_t))$ and stepsize $\eta \leq 1/L$. Define the sets of iterations $S_{\leq C} \triangleq \{t \in \{1, \dots, T\} : \|\nabla f(x_t)\|_2 \leq C\}$ and $S_{>C} \triangleq \{1, \dots, T\} \setminus S_{\leq C}$. Then for all $T \geq 1$:*

$$(i) \text{ Monotonic descent: } f(x_{t+1}) \leq f(x_t) \quad \forall t \geq 1. \quad (1)$$

$$(ii) \text{ Large-gradient steps are finite: } |S_{>C}| \leq \frac{2(f(x_1) - f_{\inf})}{\eta C^2}. \quad (2)$$

$$(iii) \text{ Summability on small-gradient steps: } \sum_{t \in S_{\leq C}} \|\nabla f(x_t)\|_2^2 \leq \frac{2(f(x_1) - f_{\inf})}{\eta}. \quad (3)$$

$$(iv) \text{ Stationarity rate: } \min_{1 \leq t \leq T} \|\nabla f(x_t)\|_2^2 \leq \frac{2(f(x_1) - f_{\inf})}{\eta \max\{1, T - \frac{2(f(x_1) - f_{\inf})}{\eta C^2}\}}. \quad (4)$$

In particular, for all $T > \frac{2(f(x_1) - f_{\inf})}{\eta C^2}$, $\min_{1 \leq t \leq T} \|\nabla f(x_t)\|_2^2 \leq \frac{2(f(x_1) - f_{\inf})}{\eta(T - \frac{2(f(x_1) - f_{\inf})}{\eta C^2})} = \mathcal{O}(1/T)$.

Theoretical Properties

- Stability via clipping: The step direction is always bounded in norm by C , ensuring $\|x_{t+1} - x_t\|_2 \leq \eta C$. This controls the local Lipschitz error term. - Hyperparameter sensitivity: - If $\eta \leq 1/L$, monotone descent holds. - Larger C leads to fewer “large-gradient” steps but potentially larger oscillations; smaller C yields stronger regularization of steps. - Comparison to un-clipped GD: Setting $C = \infty$ recovers classical GD and the descent lemma proof gives the standard non-convex rate $\min_{t \leq T} \|\nabla f(x_t)\|_2^2 = \mathcal{O}(1/T)$ for $\eta \leq 1/L$. Clipping preserves this rate and additionally bounds the number of large-gradient excursions.

Discussion

- Relation to Clipped Adam: The proposed practical algorithm uses momentum and second-moment adaptivity with clipping applied to raw gradients. The above theorem establishes a baseline non-asymptotic guarantee for the special case without momentum/adaptivity. This covers the core effect of clipping and smoothness. Extensions to momentum and Adam-style preconditioning typically introduce multiplicative constants depending on β_1, β_2 and require potential-function arguments; these do not change the asymptotic $\mathcal{O}(1/T)$ rate for the minimal gradient-norm squared under standard small-stepsize conditions. - Deterministic vs. stochastic: The above analysis is deterministic. In stochastic settings, expectations can be applied throughout. With clipping, one also

controls heavy-tailed noise; the expectation effectively replaces each occurrence of $f(x_t)$ and $\nabla f(x_t)$ by their expectations, and the same inequalities hold in expectation because the proof is linear in these quantities.

Preliminaries (Definitions and Lemmas)

Definition 1 (Clipping factor). For $g \in \mathbb{R}^d$ and $C > 0$, define the scalar clipping factor

$$s(g) \triangleq \min \left\{ 1, \frac{C}{\|g\|_2} \right\} \in (0, 1],$$

and the clipped gradient $u = \text{clip}_C(g) = s(g)g$.

Lemma 1 (Smoothness descent lemma). Under Assumption 1, for any $x, y \in \mathbb{R}^d$,

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|_2^2.$$

Proof:

[Step 1] Define $\phi(\tau) = f(x + \tau(y - x))$, $\tau \in [0, 1]$. Then $\phi'(0) = \langle \nabla f(x), y - x \rangle$.

[Step 2] By the fundamental theorem of calculus, $f(y) - f(x) = \int_0^1 \phi'(\tau) d\tau$.

[Step 3] $\phi'(\tau) = \langle \nabla f(x + \tau(y - x)), y - x \rangle$.

[Step 4] $\phi'(\tau) = \langle \nabla f(x), y - x \rangle + \langle \nabla f(x + \tau(y - x)) - \nabla f(x), y - x \rangle$.

[Step 5] Take absolute value of the second term and use Cauchy-Schwarz:

$$|\langle \nabla f(x + \tau(y - x)) - \nabla f(x), y - x \rangle| \leq \|\nabla f(x + \tau(y - x)) - \nabla f(x)\|_2 \cdot \|y - x\|_2.$$

[Step 6] Apply L -smoothness: $\|\nabla f(x + \tau(y - x)) - \nabla f(x)\|_2 \leq L\tau \|y - x\|_2$.

[Step 7] $\Rightarrow \phi'(\tau) \leq \langle \nabla f(x), y - x \rangle + L\tau \|y - x\|_2^2$.

[Step 8] Integrate over $\tau \in [0, 1]$: $f(y) - f(x) \leq \langle \nabla f(x), y - x \rangle + \int_0^1 L\tau \|y - x\|_2^2 d\tau$.

[Step 9] $\int_0^1 L\tau d\tau = L/2 \Rightarrow f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|_2^2$.

Lemma 2 (Properties of norm clipping). For $g \in \mathbb{R}^d$, $u = \text{clip}_C(g) = s(g)g$ with $s(g) = \min\{1, C/\|g\|_2\}$:

$$[\text{Step 10}] \quad \|u\|_2 \leq C. \tag{5}$$

$$[\text{Step 11}] \quad \langle g, u \rangle = s(g) \|g\|_2^2 = \min\{\|g\|_2, C\} \|g\|_2. \tag{6}$$

$$[\text{Step 12}] \quad \|u\|_2^2 = s(g)^2 \|g\|_2^2 = \min\{\|g\|_2^2, C^2\}. \tag{7}$$

Proof: Immediate from $u = s(g)g$ and the definition of $s(g)$. For (5), $\|u\|_2 = s(g)\|g\|_2 \leq C$ by the choice of $s(g)$. For (6), $\langle g, u \rangle = s(g)\langle g, g \rangle = s(g)\|g\|_2^2$. For (7), $\|u\|_2^2 = s(g)^2\|g\|_2^2$. $\square$

Main Proof (Step-by-Step Derivation)

We prove Theorem 1.

Proof: Let $g_t = \nabla f(x_t)$ and $u_t = \text{clip}_C(g_t)$. Consider the CGD update $x_{t+1} = x_t - \eta u_t$ with $\eta \leq 1/L$.

One-step descent. Apply Lemma 1 with $x = x_t$ and $y = x_{t+1} = x_t - \eta u_t$:

$$\begin{aligned} \text{[Step 13]} \quad f(x_{t+1}) &\leq f(x_t) + \langle \nabla f(x_t), -\eta u_t \rangle + \frac{L}{2} \|\eta u_t\|_2^2. \\ \text{[Step 14]} \quad &= f(x_t) - \eta \langle g_t, u_t \rangle + \frac{L\eta^2}{2} \|u_t\|_2^2. \end{aligned} \tag{8}$$

By Lemma 2, (6) and (7):

$$\text{[Step 15]} \quad \langle g_t, u_t \rangle = \min\{\|g_t\|_2, C\} \|g_t\|_2, \tag{9}$$

$$\text{[Step 16]} \quad \|u_t\|_2^2 = \min\{\|g_t\|_2^2, C^2\}. \tag{10}$$

Plugging these identities into (8) yields

$$\text{[Step 17]} \quad f(x_{t+1}) \leq f(x_t) - \eta \min\{\|g_t\|_2, C\} \|g_t\|_2 + \frac{L\eta^2}{2} \min\{\|g_t\|_2^2, C^2\}. \tag{11}$$

Case analysis. Define $A_t \triangleq \{\|g_t\|_2 \leq C\}$ and $B_t \triangleq \{\|g_t\|_2 > C\}$. - If A_t holds, then $\min\{\|g_t\|_2, C\} = \|g_t\|_2$ and $\min\{\|g_t\|_2^2, C^2\} = \|g_t\|_2^2$, so

$$\begin{aligned} \text{[Step 18]} \quad f(x_{t+1}) &\leq f(x_t) - \eta \|g_t\|_2^2 + \frac{L\eta^2}{2} \|g_t\|_2^2 \\ \text{[Step 19]} \quad &= f(x_t) - \eta \left(1 - \frac{L\eta}{2}\right) \|g_t\|_2^2 \\ \text{[Step 20]} \quad &\leq f(x_t) - \frac{\eta}{2} \|g_t\|_2^2 \quad \text{since } \eta \leq \frac{1}{L} \Rightarrow 1 - \frac{L\eta}{2} \geq \frac{1}{2}. \end{aligned} \tag{12}$$

- If B_t holds, then $\min\{\|g_t\|_2, C\} = C$ and $\min\{\|g_t\|_2^2, C^2\} = C^2$, so

$$\begin{aligned} \text{[Step 21]} \quad f(x_{t+1}) &\leq f(x_t) - \eta C \|g_t\|_2 + \frac{L\eta^2}{2} C^2 \\ \text{[Step 22]} \quad &\leq f(x_t) - \eta C^2 + \frac{L\eta^2}{2} C^2 \quad \text{since } \|g_t\|_2 > C \Rightarrow C \|g_t\|_2 \geq C^2 \\ \text{[Step 23]} \quad &= f(x_t) - \eta C^2 \left(1 - \frac{L\eta}{2}\right) \\ \text{[Step 24]} \quad &\leq f(x_t) - \frac{\eta C^2}{2} \quad \text{since } \eta \leq \frac{1}{L}. \end{aligned} \tag{13}$$

Monotonicity. Combining (12) and (13) shows $f(x_{t+1}) \leq f(x_t)$ for all t [Step 25], proving (1).

Bounding the number of large-gradient steps. Summing (13) over $t \in S_{>C}$,

$$\begin{aligned} \text{[Step 26]} \quad f(x_{T+1}) &\leq f(x_1) - \sum_{t \in S_{>C}} \frac{\eta C^2}{2} \\ \text{[Step 27]} \quad &\leq f_{\inf} \quad (\text{since } f \text{ is bounded below by } f_{\inf}). \end{aligned}$$

Rearranging,

$$[\text{Step 28}] \quad \frac{\eta C^2}{2} |S_{>C}| \leq f(x_1) - f_{\inf} \Rightarrow |S_{>C}| \leq \frac{2(f(x_1) - f_{\inf})}{\eta C^2}, \quad (14)$$

which is (2).

Summability of small-gradient steps. Summing (12) over $t \in S_{\leq C}$ gives

$$[\text{Step 29}] \quad f(x_{T+1}) \leq f(x_1) - \sum_{t \in S_{\leq C}} \frac{\eta}{2} \|g_t\|_2^2$$

$$[\text{Step 30}] \quad \leq f_{\inf}$$

and thus

$$[\text{Step 31}] \quad \sum_{t \in S_{\leq C}} \|g_t\|_2^2 \leq \frac{2(f(x_1) - f_{\inf})}{\eta}, \quad (15)$$

which is (3).

Stationarity rate. If $T \leq |S_{>C}|$, the bound (2) already limits how long the large-gradient regime can persist [Step 32]. If $T > |S_{>C}|$, then $|S_{\leq C}| \geq T - |S_{>C}|$ and

$$\begin{aligned} [\text{Step 33}] \quad \min_{1 \leq t \leq T} \|g_t\|_2^2 &\leq \frac{1}{|S_{\leq C}|} \sum_{t \in S_{\leq C}} \|g_t\|_2^2 \\ [\text{Step 34}] \quad &\leq \frac{2(f(x_1) - f_{\inf})}{\eta |S_{\leq C}|} \leq \frac{2(f(x_1) - f_{\inf})}{\eta \left(T - \frac{2(f(x_1) - f_{\inf})}{\eta C^2}\right)}, \end{aligned} \quad (16)$$

using (15) and (14) in [Step 34]. This proves (4). $\square$

Rate Derivation (With Constants)

From (16), for any $T > \frac{2(f(x_1) - f_{\inf})}{\eta C^2}$,

$$\min_{1 \leq t \leq T} \|\nabla f(x_t)\|_2^2 \leq \frac{2(f(x_1) - f_{\inf})}{\eta \left(T - \frac{2(f(x_1) - f_{\inf})}{\eta C^2}\right)} \leq \frac{4(f(x_1) - f_{\inf})}{\eta T},$$

where the last inequality uses $T - a \geq T/2$ provided $T \geq 2a$ [Step 35], with $a = \frac{2(f(x_1) - f_{\inf})}{\eta C^2}$. Thus, an explicit constant-rate bound is

$$\min_{1 \leq t \leq T} \|\nabla f(x_t)\|_2^2 \leq \frac{4(f(x_1) - f_{\inf})}{\eta} \cdot \frac{1}{T} \quad \text{for all } T \geq \frac{4(f(x_1) - f_{\inf})}{\eta C^2}.$$

Special Cases

- No clipping ($C \rightarrow \infty$): Then $u_t = \nabla f(x_t)$, and (11) with the A_t -case alone yields $f(x_{t+1}) \leq f(x_t) - \eta(1 - \frac{L\eta}{2}) \|\nabla f(x_t)\|_2^2$. For $\eta \leq 1/L$, $\sum_{t=1}^T \|\nabla f(x_t)\|_2^2 \leq \frac{2(f(x_1) - f_{\inf})}{\eta}$, implying $\min_{1 \leq t \leq T} \|\nabla f(x_t)\|_2^2 \leq$

$\frac{2(f(x_1) - f_{\inf})}{\eta T}$. - With momentum and adaptivity (Clipped Adam): The practical algorithm includes m_t, v_t and bias-corrections. Although the full proof requires additional potential-function arguments and coordinate-wise bookkeeping, the key descent mechanism remains governed by the clipped direction u_t . Under sufficiently small effective stepsizes (componentwise), analogous inequalities to (12)–(13) hold with constants scaled by factors depending on $(1 - \beta_1)^{-1}$ and the preconditioner bounds induced by ϵ and the second moment. The asymptotic $\mathcal{O}(1/T)$ rate for $\min_{t \leq T} \|\nabla f(x_t)\|_2^2$ is preserved up to multiplicative constants.